Econometrics I

Linear Algebra Review (Self-Study)

Fall 2026

How to use this handout

- Read this before Lecture 2. It is the linear algebra we rely on all semester — nothing more.
- You will *read* matrix notation constantly in this course; you will almost never manipulate it by hand.
- If any of this is new to you, work through the linked visualizations and come to office hours early in the semester.

- A **vector** in $\mathbb{R}^n$ is a column of numbers $(x_1, x_2, ..., x_n)$.
- A **matrix** in $\mathbb{R}^{n \times m}$ is m columns of length n vectors (so n is the number of rows and m is the number of columns). We denote an element of a matrix by m_{ij} for row i and column j:

$$M = \begin{pmatrix} m_{11} & m_{12} \\ m_{21} & m_{22} \end{pmatrix}$$

- For an entity on which there are many pieces of data, we store the data in a vector x_i

  Example: For the USA could have $x_{USA} = (GDP_{USA}, Population_{USA}, ...)$
- For many entities we can store all the data in a data matrix, X.

  Example: For two countries:

$$X = \begin{pmatrix} GDP_{USA} & Population_{USA} \\ GDP_{Canada} & Population_{Canada} \end{pmatrix}$$

Matrix Multiplication

- For two vectors of equal length define the **dot product** as $v \cdot w$ or $\langle v, w \rangle$:

$$v \cdot w = \sum_{i=1}^{n} v_i \times w_i$$

- For two matrices, A and B of sizes $n \times m$ and $m \times k$ define the **matrix product** $C = AB$ as the $n \times k$ matrix with entries $c_{ij} = \sum_{l=1}^{m} a_{il} b_{lj}$
    - Easy way to remember: $(i, j)^{th}$ element of product is dot product of i^{th} row and j^{th} column of A and B respectively.
    - Not all matrices can be multiplied: left matrix must have column length equal to right matrix's row length
    - Multiplication is NOT commutative: $AB \neq BA$ even if they both exist

https://eli.thegreenplace.net/2015/visualizing-matrix-multiplication-as-a-linear-combination/

Transposes

- Define the **transpose** of A as the matrix A' with elements $a'_{ij} = a_{ji}$ (reverse columns and rows)
- A matrix is **symmetric** if $A' = A$
- **Important Properties:**
  - The matrix $B = A'A$ is *always* a square matrix
  - The matrix $B = A'A$ is always symmetric
  - $(A')' = A$
  - **Multiplication Rule:** $(AB)' = B'A'$
  - **Addition Rule:** $(A + B)' = A' + B'$
- Note that dot products can also be written as an **inner product**: $v \cdot w = v'w$.
- Note that the **outer product** of two vectors is a matrix rather than a scalar: vw'.

- The **Identity Matrix**, I, is a matrix with 1s on the diagonal and 0s elsewhere. Clearly $AI = A$.
- Define the **left inverse** of A to be the matrix A^{-1} such that $A^{-1}A = I$
  - Can analogously define right inverse
  - Right and left inverse will NOT be the same if A is not a square matrix
  - Right and left inverse WILL be equal if A is square (then we just say inverse)
- **Important Properties:**
  - **Multiplication Rule:** $(AB)^{-1} = B^{-1}A^{-1}$
  - **Tranpose Rule:** $(A')^{-1} = (A^{-1})'$
  - **Dot Product:** $v \cdot w = v'w$

- For a function $f : \mathbb{R}^n \to \mathbb{R}^m$ recall the definition of the derivative or Jacobian of f:

$$Df = \begin{pmatrix} \frac{\partial f_1}{\partial x_1} & \dots & \frac{\partial f_m}{x_1} \\ \vdots & & \\ \frac{\partial f_1}{\partial x_n} & \dots & \frac{\partial f_m}{x_n} \end{pmatrix}$$

- We WON'T be doing anything too complicated! But we can define two important functions given a vector x and a matrix A:
  - For Ax, $D(Ax) = A$ (as a line in 1-D calc)
  - For $x'Ax$, $D(x'Ax) = x'(A + A')$ (as a quadratic in 1-D calc)

The matrix cookbook `https://www.math.uwaterloo.ca/~hwolkowi/matrixcookbook.pdf` has a lot (more) helpful properties.

Check yourself

Before Lecture 2, make sure you can answer:

1. If X is $n \times k$, what are the dimensions of $X'X$? Of $X'y$ (with y being $n \times 1$)? Of $(X'X)^{-1}X'y$?
2. Why is $X'X$ always symmetric?
3. When does $(X'X)^{-1}$ fail to exist? Give a two-column example.
4. Expand $(y - X\beta)'(y - X\beta)$ using the transpose rules.